

Sessional Test-II

Dated: 18-11-2024, Full Marks: 30

Course Name: Physics, Course Code: BPY23111T

Programme: B. Tech In Civil Engineering

1. Fill in the blanks: (1x5=5)

- (i) The aperture is used to block the _____
- (ii) The distance between the lenses of focal lengths, f_1 and f_2 minimize spherical aberration is _____
- (iii) The wavelength emitted from He-Ne laser is _____
- (iv) The total number of Einstein coefficients is _____
- (v) Coherent photons are emitted in _____ emission.

2. Answer the following questions briefly (2x5=10)

- (i) What do you mean by spontaneous emission.
- (ii) Explain in brief the use of stop in spherical aberration.
- (iii) What is achromatic doublet?
- (iv) Define population inversion.
- (v) Draw the diagram of the construction of Nd-YAG laser.

3. Answer any one from the following: (5)

- (i) Proof that measure of chromatic aberration is given by the product of dispersive power and focal length.
- (ii) Explain the Einstein's coefficients..

4. Answer any one of the following (10)

- (i) Deduce the expression of achromatism in case of two lenses separated by some distance.
- (ii) Explain the working principle of He- Ne laser.